

Junhyeong Lee

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Research Interests

Artificial intelligence for inverse design · Artificial intelligence for manufacturing · Data-driven optimization · Large Language Models

Education & Experience

Postdoctoral Fellow Feb. 2026 – Present
PRISM-AI Center (InnoCORE) & AiM⁴ Lab, Korea Advanced Institute of Science and Technology (KAIST), S. Korea

Advisor: Prof. Seunghwa Ryu

Ph.D. in Mechanical Engineering Sep. 2022 – Feb. 2026

Korea Advanced Institute of Science and Technology (KAIST), S. Korea

Advisor: Prof. Seunghwa Ryu

Research Assistant Mar. 2022 – Aug. 2022

Department of Mechanical Engineering, KAIST, S. Korea

Advisor: Prof. Seunghwa Ryu

M.S. in Mechanical Engineering Mar. 2020 – Feb. 2022

Korea Advanced Institute of Science and Technology (KAIST), S. Korea

Advisor: Prof. Seunghwa Ryu

B.Eng. in Convergence Engineering, *Summa Cum Laude* Mar. 2015 – Feb. 2019

Daegu Gyeongbuk Institute of Science and Technology (DGIST), S. Korea

Publications

⁺ *Equal contribution* ^{*} *Corresponding author*

2026

14. Cho, H., Yu, J., Moon, H., Yoon, J., **Lee, J.**, Kim, G., Park, J., & Ryu, S.* “Real-Time Structural Health Monitoring with Bayesian Neural Networks: Distinguishing Aleatoric and Epistemic Uncertainty for Digital Twin Frameworks.” **International Journal of Precision Engineering and Manufacturing – Green Technology**, accepted. [SSRN]
13. Lee, S., **Lee, J.**, Park, D., Lee, S., Lee, K.* , Kim, H.* , & Ryu, S.* “Reducing False Negatives in Gastroesophageal Reflux Disease Diagnosis Through Multi-Feature Anomaly Detection of pH-Impedance Signals.” **npj Digital Medicine**, In press.
12. **Lee, J.**⁺, Kim, J.⁺, Kee, S.⁺, Yu, J., & Ryu, S.* “Federated Learning Framework for Data-Sovereign Quality Prediction in Injection Molding.” **International Journal of Precision Engineering and Manufacturing**, In press. [10.1007/s12541-026-01546-0](https://doi.org/10.1007/s12541-026-01546-0)

2025

11. **Lee, J.**⁺, Kim, J.⁺, Kim, H.⁺, Lee, I., & Ryu, S.* “IM-Chat: A Multi-agent LLM-based Framework for Knowledge Transfer in Injection Molding Industry.” **Journal of Manufacturing Systems** (IF: 14.2, JCR Top 0.7%), In press. [10.1016/j.jmsy.2025.11.007](https://doi.org/10.1016/j.jmsy.2025.11.007)

10. Lee, H.⁺, Moon, H.⁺, **Lee, J.⁺**, & Ryu, S.* “Toward Knowledge-Guided AI for Inverse Design in Manufacturing: A Perspective on Domain, Physics, and Human-AI Synergy.” **Advanced Intelligent Discovery**, e202500107. [10.1002/aidi.202500107](https://doi.org/10.1002/aidi.202500107)
9. Kang, H.⁺, **Lee, J.⁺**, Lee, S., Jeon, J., Mun, C., Yang, J., Kwon, H., Lee, I., Kim, S., Lim, E., Jung, J., Jung, Y.* , Park, S.* , Ryu, S.* , & Kang, T.* “Interpretability-Driven Deep Learning for SERS-based Classification of Respiratory Viruses.” **Biosensors and Bioelectronics**, 117891 (IF: 10.5, JCR Top 3.2%). [10.1016/j.bios.2025.117891](https://doi.org/10.1016/j.bios.2025.117891)
8. Yoon, J.⁺, **Lee, J.⁺**, Ryu, S.* , & Park, J.* “Enhanced Energy Harvesting in Rotary Triboelectric Nanogenerators via Gaussian Process Regression-Based Bayesian Optimization.” **Nano Energy**, 135, 110653 (IF: 17.1, JCR Top 7.8%). [10.1016/j.nanoen.2025.110653](https://doi.org/10.1016/j.nanoen.2025.110653)
7. Moon, H., **Lee, J.**, Yu, J., & Ryu, S.* “Physics-Informed AI for Material Characterization: A Perspective on Data-Efficient Discovery through Physics-Informed Neural Networks.” **International Journal of AI for Materials and Design**, 2(4). [10.36922/IJAMD025440043](https://doi.org/10.36922/IJAMD025440043)

2024

6. **Lee, J.**, Park, K., Park, K., Kim, Y., Kim, J.* , & Ryu, S.* “Electrode Placement Optimization for Electrical Impedance Tomography using Active Learning.” **Advanced Engineering Materials**, 26(11), 2301865. [10.1002/adem.202301865](https://doi.org/10.1002/adem.202301865)
5. Kang, H.⁺, **Lee, J.⁺**, Moon, J.⁺, Lee, T.⁺, Lim, E., Jung, J., Jung, Y., Lee, S., Lee, K.* , Ryu, S.* , & Kang, T.* “Multiplex Detection of Foodborne Pathogens using 3D Nanostructure Swab and Deep Learning-based Classification Raman Spectra.” **Small**, 20(35), 2308317 (IF: 13.0, JCR Top 10.6%). (front cover) [10.1002/smll.202308317](https://doi.org/10.1002/smll.202308317)

2023

4. **Lee, J.**, Park, D., Lee, M., Lee, H., Park, K., Lee, I., & Ryu, S.* “Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: a review.” **Materials Horizons**, 10, 5436–5456 (IF: 13.3, JCR Top 9.8%). [10.1039/D3MH00039G](https://doi.org/10.1039/D3MH00039G)

2022

3. Yoon, J.⁺, **Lee, J.⁺**, Kim, G., Ryu, S.* , & Park, J.* “Deep neural network-based structural health monitoring technique for real-time crack detection and localization using strain gauge sensors.” **Scientific Reports**, 12(1), 20204. [10.1038/s41598-022-24269-4](https://doi.org/10.1038/s41598-022-24269-4)

2021

2. Park, K., **Lee, J.**, & Ryu, S.* “Damage detection of composite materials via IR thermography and electrical resistance measurement: A review.” **Structural Engineering and Mechanics**, 80(5), 563–583. [10.12989/sem.2021.80.5.563](https://doi.org/10.12989/sem.2021.80.5.563)

2020

1. Lim, S. H.⁺, Sohn, S. W.⁺, Lee, H., Choi, D., Jang, E., Kim, M., **Lee, J.**, & Park, S.* “Analysis and evaluation of path planning algorithms for autonomous driving of electromagnetically actuated microrobot.” **International Journal of Control, Automation and Systems**, 18, 2943–2954. [10.1007/s12555-019-0637-9](https://doi.org/10.1007/s12555-019-0637-9)

Under Review / Submitted

1. **Lee, J.**, Kim, J.-Y., & Ryu, S.* “Before You Build a Multi-Agent System: An Escalation Framework for LLM Adaptation.” Submitted. [[engrXiv](https://arxiv.org/abs/2505.14201)]
2. Yu, J.⁺, Byun, J.⁺, Moon, H., **Lee, J.**, & Ryu, S.* “Dual-Constrained Diffusion Model for Nonlinear Microstructure Inverse Design with Direct Evaluation.” Under review.
3. Demeke, W.⁺, **Lee, J.⁺**, Lee, I.⁺, Ryu, B.* , & Ryu, S.* “A Comprehensive Review of Thermoelectric Power Generators: Advances in Material Discovery, Structural Optimization for Performance Enhancement, and Service Life Stability.” Under review.
4. Lee, I.⁺, **Lee, J.⁺**, Park, J., Lim, K.* , & Ryu, S.* “Enhancing LLM-based Conditional Material Generation for Double Perovskite through Multi-Agent Feedback and Domain Knowledge.”

Patents

1. Kim, J., **Lee, J.**, Kim, H., Lee, I., Ryu, S., Nam, K., & Yoon, S. (2025). “Knowledge Transfer Device and Method using an Agent Based on a Large Language Model.” Filed (KR 10-2025-0145843).
2. Ryu, S., **Lee, J.**, & Park, K. (2023). “Method and Apparatus for Optimization of Cathode Material Design Multi-objective Bayesian Optimization.” Filed (KR 10-2023-0172701).

Conferences

1. Choi, D., Jang, E., Kim, M., **Lee, J.**, Lim, S., Sohn, S., Lee, H., & Park, S.* (2018). “Analysis of Path Planning Algorithm for Autonomous Control of Electromagnetically Actuated Microrobot.” **SMIT2018-IBEC2018 Joint Conference** (Poster).
2. **Lee, J.**, & Ryu, S.* (2021). “Data-Driven Electrode Placement Optimization for Crack Sensing of Carbon-based Composites using Electrical Resistance Tomography.” **KSME CAE & Applied Mechanics Conference** (Oral).
3. **Lee, J.**, Lee, H., & Ryu, S.* (2022). “Electrode Placement Optimization for Electrical Resistance Tomography using Transfer Learning.” **KSME CAE & Applied Mechanics Conference** (Oral).
4. **Lee, J.**, & Ryu, S.* (2022). “Deep Learning-based Electrode Placement Optimization for Crack Sensing of Carbon-based Composites using Electrode Resistance Tomography.” **GRC Additive Manufacturing of Soft Materials** (Poster).
5. **Lee, J.**⁺, Yoon, J.⁺, Kim, G., Park, J.* & Ryu, S.* (2023). “A Deep Learning Approach for Crack Sensing in Structural Health Monitoring using Strain Gauge Sensors.” **COSEIK** (Oral).
6. **Lee, J.**⁺, Yoon, J.⁺, Kim, G., Park, J.* & Ryu, S.* (2023). “Real-time Crack Localization using Strain Gauge Sensors with Deep Learning.” **KSME CAE & Applied Mechanics Conference** (Oral).
7. **Lee, J.**⁺, Park, D., Lee, M., Park, K., Lee, H., Lee, I., Lee, I., & Ryu, S.* (2024). “Review of Machine Learning-based Inverse Design Methods in Materials Design and Manufacturing.” **KSME CAE & Applied Mechanics Conference** (Poster).
8. Kang, H.⁺, **Lee, J.**⁺, Moon, J.⁺, Lee, T.⁺, Lim, E., Jung, J., Jung, Y., Lee, S., Lee, K.* & Ryu, S.* & Kang, T.* (2024). “Detection of Multiple Foodborne Pathogens Utilizing a Three-dimensional Nanostructure Swab and Deep Learning-based Analysis of Raman Spectra.” **Nano Korea** (Oral).
9. Kang, H.⁺, **Lee, J.**⁺, Moon, J.⁺, Lee, T.⁺, Lim, E., Jung, J., Jung, Y., Lee, S., Lee, K.* & Ryu, S.* & Kang, T.* (2024). “3D Nanostructure Swab and Deep Learning Analysis of Raman Spectra for Detection of Multiple Foodborne Pathogens.” **KSME Conference** (Oral).
10. Lee, I., **Lee, J.**, & Ryu, S.* (2025). “Multi-Agent Argumentative LLM Framework for Conditional Exploration of Double-Perovskite Materials.” **Spring Conference of the Korean Institute of Metals and Materials** (Oral).
11. **Lee, J.**⁺, Kim, J.⁺, Kim, H.⁺, Lee, I., Park, J., & Ryu, S.* (2025). “Enhancing Knowledge Transfer in Injection Molding Through a Multi-Agent LLM Framework.” **KSME CAE & Applied Mechanics Conference** (Oral).
12. **Lee, J.**, Kim, J., Kim, H., Lee, I., Choi, H., & Ryu, S.* (2026). “LLM-based Knowledge Transfer System for Injection Molding: Implementation and Performance Enhancement.” **KSME CAE & Applied Mechanics Conference** (Oral).

Honors & Awards

Excellent Dissertation Award, KSME Headquarters	2026
Outstanding Paper Award, KSME CAE & Applied Mechanics Division	2026
Woo-Deok Future Scholar (), Woo-Deok Foundation	2025
Outstanding Paper Award, KSME CAE & Applied Mechanics Division	2025
DGIST Presidential Fellowship	2016–2019
• Placed on the Dean’s list. • \$10K expenses for Undergraduate Group Research Program • \$20K financial support for overseas lab experience	
Dean’s List, DGIST	2015
• Included in the top 5 excellent students ranking in each semester • \$2.5K financial support for undergraduate study	

Projects

InnoCORE Project, MSIT Korea	2025–Present
• National flagship program (~\$25M USD over 4.5 years) to advance AI-driven manufacturing across KAIST, GIST, DGIST, and UNIST. • Led proposal development and conceptual design of an LLM-based autonomous manufacturing framework. • Developing LLM-driven methods for materials design, structural optimization, and process parameter tuning.	
National Agenda Basic Research Program, MSIT Korea	2025–Present
• Three-year project (~\$150K USD/year) developing domain-specialized LLMs for advanced manufacturing productivity. • Contributing to core LLM frameworks enabling intelligent, domain-aware automation in smart manufacturing.	

Teaching & Leadership

AIM⁴ Lab Student Representative, KAIST	2024
• Coordinated research activities, organized internal seminars, managed lab resources	
Teaching Assistant, KAIST	2022–2025
• ME567: Introduction to Statistical Thermodynamics (2022) • ME231: Solid Mechanics (2023, 2025) • ME639: Elasticity & Micromechanics (2023) • ME211: Thermodynamics (2024) • ME491B: Statistical Data Analysis (2024) • ME221: Fluid Mechanics (2024, 2025)	
Teaching Assistant, DGIST	2019
• SE201b: Linear Algebra (2019)	

Skills

Simulation / Coding: COMSOL Multiphysics, ABAQUS, Fe-safe, HyperMesh, ADINA, EIDORS,

MATLAB, Python

Data-Driven Design: Generative modeling, surrogate modeling & optimization, Bayesian optimization

Large Language Model: Multi-agent orchestration, RAG (retrieval-augmented generation), tool-calling

Languages: Korean (native), English (advanced)

References

Prof. Seunghwa Ryu Department of Mechanical Engineering, KAIST ryush@kaist.ac.kr

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